

# Nyx Haile

## Quantitative Developer

xyn@mit.edu | LinkedIn | GitHub | Open to relocation

### EDUCATION

#### Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics, Minor in Economics

Cambridge, MA

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

### EXPERIENCE

#### Partial

2025 – Present

Cofounder & CEO

London

- Designed a perpetual prediction-market exchange where positions are probability distributions over joint price paths across correlated assets, not binary outcomes.
- Specified eleven market types, among them price paths, derived statistics (correlations, lead-lag) and reported statistics, under a single mechanism-design framework.
- Restored strict 0.02 forward-KL / 0.02 relative-price-error projection thresholds as the engine's fail-closed runtime default, keeping looser bounds for diagnostics only; 12 open model failures remain tracked.

### SELECTED PROJECTS

Repositories marked private are available on request.

#### MIT Pokerbots | Python

Heads-up poker-variant bot, January 2026

- Built a Python bot for MIT Pokerbots' heads-up poker variant, largely with AI-assisted coding; the team placed about 15th of 70.
- Implemented Monte Carlo equity estimation, with exact enumeration for discards, using quick and deep sampling paths to fit a per-decision compute budget.
- Built a multi-policy architecture (tight, bluff, desperate, lock-win) with a strategic overseer choosing among threshold policies.

#### howtown.lol | Python, PyTorch, CUDA, Redis

World model for post-game teaching

- Designed a Dreamer-lineage RSSM world model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.

#### Strand (private) | Rust, FoundationDB

Capability-authenticated issue tracker

- Wrote stable-Rust fuzz harnesses for three frozen v1 formats (wire frames, CBOR citations, backup files); the first pass ran 284,846 iterations with zero crashes.

#### Neuro (private) | Python, NumPy

Exploratory adaptive sparse neural core

- Implemented temperature-controlled sampled activation, copula-driven rewiring, a loss-plus-sparsity objective and automatic pruning with edge rebalancing.
- Wrote a research note relating the model to resource-bounded, stream-based approximations of Solomonoff induction, and where it falls short of them.

#### Forge (private) | Rust, systemd, RPC

Local-first agent control plane

- Added bounded, jittered exponential backoff to scheduler dispatch, seeded per task so retry waves de-correlate in production and replay exactly in tests.

### TECHNICAL SKILLS

**Languages:** Rust, Python, TypeScript, SQL   **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval   **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis